

Elisa CUEILLE

PhD Student in Computer Science

Contact

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Research Experience

PhD in Computer Science : October 2024-2027

Title : Agent behavior generation under environmental constraints.

University : Toulouse Capitole University

Laboratory : Toulouse Institute of Computer Science Research (**IRIT**)

Team : Cooperative Multi-Agent Systems (**SMAC**)

PhD Advisors : Benoit GAUDOU, Nicolas VERSTAEVEL, Manon PRÉDHUMEAU

Ministerial doctoral contract

Description : The main objective is to develop a new simulation model where individuals' activity chains are no longer merely inferred from past data, but are generated based on the opportunities and constraints (physical or social) offered by a dynamic environment. Specifically, the goal is to propose a multimodal mobility model where agents adapt their decisions and movements according to the opportunities and constraints of the environment they perceive. This model will allow testing the impact of future urban planning initiatives (such as low-emission zones or the construction of new infrastructure) on residents' behavior.

Publications

1. **Cueille, E.**, Gaudou, B., Verstaevael, N., & Prédhumeau, M. (2025, May). [Assessing the impact of crisis cell decisions during flash flood](#). In The 26th International Workshop on Multi-Agent-Based Simulation.

Presentations

- **Annual Synthetic Populations Workshop 2025** : Activity-based model for multimodal urban mobility simulation in Hanoi
- **Seminar at Toulouse Capitole University 2025** : Activity-based modeling for multimodal urban mobility simulation, Toulouse, FRANCE

Awards and Grants

- **MESRI Grant** (Ministry of Higher Education, Research and Innovation) funding 3 years of PhD research, awarded by competitive examination
- **DAR Mobility Grant** (Research Support Department) of Toulouse Capitole University : €1,586

- **EDMITT Mobility Grant** (Doctoral School of Mathematics, Computer Science and Telecommunications of Toulouse) : €900

Scientific Activities

Contributions

- **GAMA Coding Camp** : Avril 2026, Ninh Binh, Vietnam
 - Participation in the development of the GAMA multi-agent simulation platform, alongside the team of researchers and developers creating the tool.
- **Representative of students and PhD candidates at the Computer Science Faculty of Toulouse Capitole University** : 2025 - Present
 - Participation in faculty meetings
- **Representative of PhD candidates in the ICI department** : 2025 - Present
 - Participation in department meetings
- **GAMA Coding Camp** : Mai 2025, Ninh Binh, Vietnam
 - Participation in the development of the GAMA multi-agent simulation platform, alongside the team of researchers and developers creating the tool.
- **Organization of scientific seminars** : 2024 - Present
 - Co-organization of IRIT seminars (Toulouse Capitole University site), aimed at fostering scientific exchange and enabling work presentations by researchers from inside and outside the laboratory.
- **Participation in the HRS4R working group** : 2024
 - Member of the "Computer Science" working group as part of the European HRS4R (Human Resources Strategy for Researchers) certification for Toulouse Capitole University.

Scientific Events

- **Games and Artificial Intelligence Multidisciplinary Summer School 2025** : June 2025, Toulouse
 - Participation in the multidisciplinary summer school focused on the intersections between game theory, mathematics, economics, and their applications across various domains.
- **eX Modelo School (OpenMole)** : December 2024, Paris
 - Training in the use of the OpenMole platform, a software tool dedicated to exploring numerical models (sensitivity analysis, calibration, optimization) and running simulations on distributed computing infrastructures.

Teaching Experience

Teaching Assistant (TD) : 2024-Present

- **Courses taught :**
 - **Entity-Relationship Model and Business Process Model and Notation**
 - **Master 1 MIAGE** at Toulouse Capitole University
 - 48 hours
 - **Python Programming**
 - **Bachelor 2nd Year Economics** at **Toulouse School of Economics**

- 67.5 hours
- **Algorithms in Python**
 - **Bachelor** 1st Year Economics and Mathematics at **Toulouse School of Economics**
 - 30 hours

Education and Academic Qualifications

Master MIAGE (Mathematics and Computer Science Applied to Human and Social Sciences) : Business Process Engineering : 2022-2024

Toulouse Capitole University

Master 2 result : Highest Honors (Mention Très Bien), Valedictorian **1/44**

Master 1 result : Highest Honors (Mention Très Bien), Valedictorian **1/97**

- Algorithms and programming
- IS Conception and Enterprise Architecture
- Modeling : BPMN, UML, UP
- Agile methodologies
- Big Data

Bachelor Degree 3rd Year MIAHS (Mathematics and Computer Science Applied to Human and Social Sciences) MIAGE track : 2021-2022

Toulouse Capitole University

Result : Highest Honors (Mention Très Bien), Valedictorian **1/76**

- Web development
- Statistics
- Algorithms and object-oriented programming
- Structured and massive unstructured data
- Financial management and project management

Bachelor Degree 1st and 2nd Year : Economics and Management : 2019-2021

University : **Toulouse School of Economics**

Bachelor 2nd year result : Highest Honors (Mention Très Bien), Valedictorian **1/463**

Bachelor 1st year result : Highest Honors (Mention Très Bien), Valedictorian **1/765**

- Mathematics
- Statistics
- Macroeconomics and Microeconomics
- Management Accounting, Financial Accounting
- Marketing
- Management

Work Experience

Work-Study Program : 2022-2024

Job Title : **Full-stack Developer and Scrum Master**

Company : **Sopra Steria**

Description :

- Two-year work-study program at Sopra Steria, divided into two distinct assignments:
 - First year (2022-2023) : **Third-Party Application Maintenance (TMA) - Banking Sector**
 - Responsibilities : Development (front-end and back-end), performance and obsolescence studies, drafting functional specifications and quotes.
 - Second year (2023-2024) : **Agile Project - Telecommunications Sector**
 - **Full-stack Developer** : Back-end and front-end development, batch job creation, and writing unit and functional tests.
 - **Scrum Master** : Facilitating daily stand-up meetings and retrospectives, implementing a burnup chart for tracking.
 - **Business Analyst / Product Owner** (ad-hoc assignments) : Writing User Stories, creating test plans, designing wireframes/mockups, process modeling (BPMN), and leading client demonstrations.

Internship : 2022

Job Title : **Full-stack Developer**

Company : **CEA Tech**

Description :

- Contribution to the **development of the DONUT tool** (Definition of Innovative Digital Tools facilitating Crisis Team Work), a **dashboard** dedicated to **crisis management for the national crisis cell at CEA**.
 - Development of a full-stack tool capable of managing and synchronizing real-time data following an agile, user-centered methodology.

Technical and Language Skills

Computer Science & Development

Programming Languages : Java, Python, JavaScript, TypeScript, React, Angular, Gaml

Frameworks & Libraries : Spring Boot, Spring Batch, Hibernate, Meteor, LoopBack

Databases : PostgreSQL, SQL, MongoDB, Microsoft SQL SMS, Access

Testing : JUnit, Mockito, Cucumber, Selenium

Tools, Platforms & DevOps : Git, Docker, Maven, GitLab CI/CD, SonarLint, IntelliJ IDEA, Eclipse, PgAdmin, Postman, SoapUI, CastleMock, GAMA

Languages

French : Native language

English : Professional proficiency